AHB-Lite DMA Controller

Register Map & Architecture Documentation

Memory-to-Memory Transfer Engine on AHB-Lite

1. Architecture Overview

This DMA controller was designed specifically for word-granularity memory-to-memory copies between two SRAMs over an AHB-Lite bus. Unlike a traditional Intel 8237-style DMA (which uses dedicated I/O-to-memory channels, 8-bit base registers, and hardware DRQ/DACK lines), this design is a software-controlled AHB peripheral. Every configuration and status bit is accessible via normal 32-bit AHB read/write transactions.

Key Design Decisions

• All registers are 32-bit wide, aligned to 4-byte boundaries.

• The control register combines start, abort, IRQ-enable, mode, and burst-length into a single word — one atomic write to configure and launch.

• Performance counters are separate read-only registers for profiling without a logic analyser.

• Separate source/destination address registers instead of an offset model.

1.1 System Architecture Block Diagram

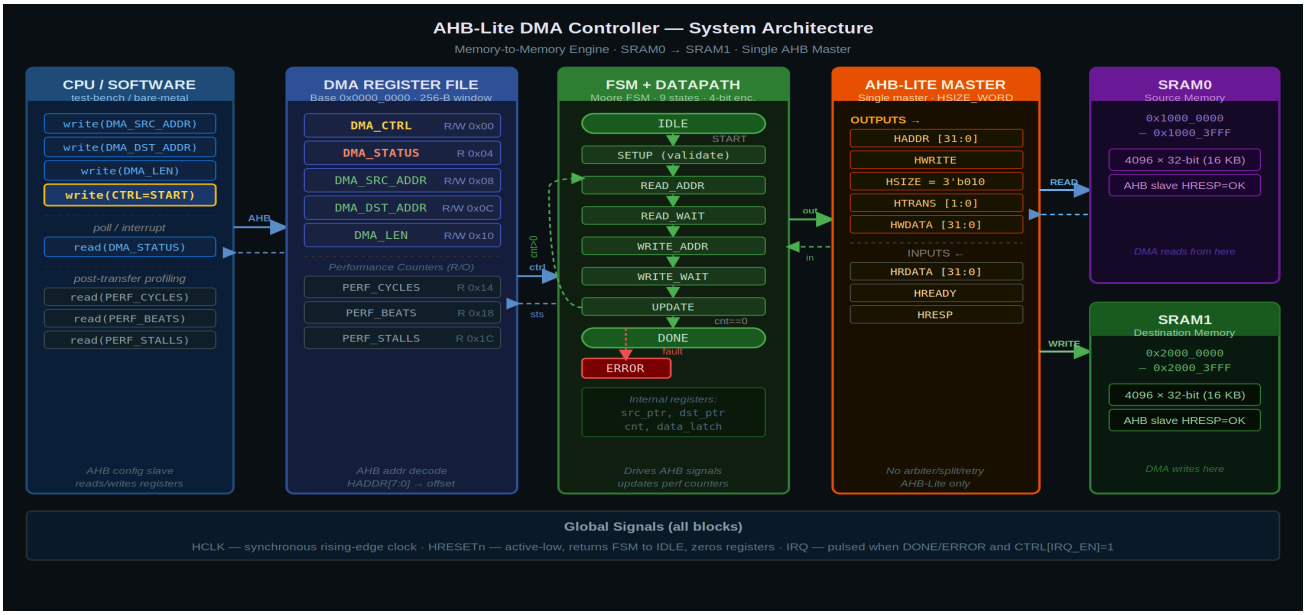

Figure 1 — AHB-Lite DMA Controller: Five-layer architecture. Left→Right: software config path. Bottom arrow: data flow SRAM0→SRAM1.

2. System Memory Map

The address space is partitioned into three non-overlapping regions. DMA registers at 0x0000_0000, SRAM0 (source) at 0x1000_0000, SRAM1 (destination) at 0x2000_0000.

The DMA register aperture is placed at 0x0000_0000 to make it easy to address from a test bench without needing a CPU MMU.Both SRAMs are placed at high addresses to give the register space room to grow.

Region	Address Range	Description
DMA Registers	**0x0000_0000 – 0x0000_00FF**	256 B aperture; 8 registers used (8 registers × 4 bytes = 32 B used; 256 B window reserved)
DMA_CTRL	0x0000_0000	Control register
DMA_STATUS	0x0000_0004	Status register
DMA_SRC_ADDR	0x0000_0008	Source address
DMA_DST_ADDR	0x0000_000C	Destination address
DMA_LEN	0x0000_0010	Transfer length
DMA_PERF_CYC	0x0000_0014	Perf counter: cycles
DMA_PERF_BEAT	0x0000_0018	Perf counter: beats
DMA_PERF_STL	0x0000_001C	Perf counter: stalls
SRAM0 (Source)	**0x1000_0000 – 0x1000_3FFF**	4096 × 32-bit = 16 KB → read source
SRAM1 (Dest)	**0x2000_0000 – 0x2000_3FFF**	4096 × 32-bit = 16 KB → write destination

The 256-byte DMA_REG_SIZE aperture (DMA_REG_BASE + 0xFF) is deliberately larger than the 8 registers actually used. This gives the design room to add features.

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3. DMA Register Map

All registers at DMA_REG_BASE = 0x0000_0000. Offsets word-aligned (4-byte steps). Unlisted bit positions within each register are Reserved and must be written as 0; reading them returns 0.

Register	Offset	Width	Purpose
DMA_CTRL	0x00	32-bit	Master control: start, abort, mode, burst length, IRQ enable

Register	Offset	Width	Purpose
DMA_STATUS	0x04	32-bit	Live status: busy, done, error flag + 4-bit error code
DMA_SRC_ADDR	0x08	32-bit	Word-aligned byte address in SRAM0
DMA_DST_ADDR	0x0C	32-bit	Word-aligned byte address in SRAM1
DMA_LEN	0x10	32-bit	Transfer length in 32-bit words (must be ≥ 1)
DMA_PERF_CYCLES	0x14	32-bit	Read-only: AHB clock cycles elapsed during last DMA run
DMA_PERF_BEATS	0x18	32-bit	Read-only: AHB data beats transferred (ideal = 2×LEN)
DMA_PERF_STALLS	0x1C	32-bit	Read-only: HREADY=0 stall cycles during last run

4. Register Bit-Field Definitions

4.1 DMA_CTRL (Offset 0x00)

Write to configure and launch. CTRL[0]=START is a write-1-pulse (self-clearing). CTRL[2]=ABORT terminates in-progress transfer.

This is the primary register for initiating and configuring a DMA transfer. Writing this register atomically sets all transfer parameters and, when bit 0 is set, launches the operation. The controller reads this register only in DMA_IDLE state; while a transfer is active, new writes are ignored (or, if the ABORT bit is set, they take effect immediately).

BURST_LEN[3:0][7:4]	MODE[3]	ABORT[2]	IRQ_EN[1]	START[0]	Reserved [31:8]

Bit(s)	Name	Access	Description
[0]	START	W1P	Write 1 to start. Self-clears. No effect while BUSY=1.
[1]	IRQ_EN	R/W	Interrupt enable. When 1, asserts IRQ on DONE or ERROR.
[2]	ABORT	R/W	Write 1 to terminate. FSM completes current beat then → DMA_ERROR(ABORT).
[3]	MODE	R/W	0 = single-beat (NONSEQ→IDLE each beat). 1 = burst (SEQ for N beats).
[7:4]	BURST_LEN	R/W	Burst count (1, 4, 8, or 16). Only meaningful when MODE=1.
[31:8]	Reserved	R(0)	Must write 0.

4.2 DMA_STATUS (Offset 0x04)

Read-only. DONE and ERROR bits are sticky until next START. Poll BUSY[0] for completion or use IRQ.

The status register is read-only from the CPU perspective. The DMA controller writes it internally as the FSM advances through states. Software should poll STATUS[BUSY] to determine when a transfer has completed, or use the interrupt mechanism if IRQ_EN was set. The DONE and ERROR bits are sticky — they hold their values until a new START pulse is issued.

ERRCODE[3:0][7:4]	Rsvd[3]	ERR[2]	DONE[1]	BUSY[0]	Reserved [31:8]

Bit(s)	Name	Access	Description
[0]	BUSY	R	Set while FSM is active. Clear when IDLE or DONE.
[1]	DONE	R	Set when FSM enters DMA_DONE. Sticky until next START.
[2]	ERROR	R	Set when FSM enters DMA_ERROR. Inspect ERRCODE[7:4].
[7:4]	ERRCODE	R	4-bit error code. Valid only when ERROR=1. See Section 5.

4.3–4.5 Address and Length Registers

DMA_SRC_ADDR (0x08): byte address in SRAM0. Bits [1:0] must be 00 (word-aligned) — violation → DMA_ERR_SRC_ALIGN.

DMA_DST_ADDR (0x0C): byte address in SRAM1. Same alignment rule — violation → DMA_ERR_DST_ALIGN.

DMA_LEN (0x10): number of 32-bit words. Must be ≥ 1 — zero value → DMA_ERR_ZERO_LEN.

4.3 DMA_SRC_ADDR — Source Address Register (Offset 0x08)

Holds the byte address of the first word to read from SRAM0. The DMA controller verifies that bits [1:0] are 00 (word-aligned) before starting. If alignment fails, the FSM immediately goes to DMA_ERROR with DMA_ERR_SRC_ALIGN without issuing any AHB transactions.

SRC_ADDR[31:2] — byte address, bits [31:2]	Must be 00[1:0]

Bit(s)	Name	Access	Description
[31:2]	SRC_ADDR	R/W	Upper 30 bits of the source byte address. Combine with implicit [1:0]=00 to form the full 32-bit AHB HADDR for the read phase.
[1:0]	Align	R/W	Must be written as 2'b00. Non-zero value triggers DMA_ERR_SRC_ALIGN and prevents the transfer from starting.

Valid range: 0x1000_0000 to 0x1000_3FFC (SRAM0 window, word-aligned, leaving room for the last word).

4.4 DMA_DST_ADDR — Destination Address Register (Offset 0x0C)

Mirror of DMA_SRC_ADDR but for SRAM1. The same alignment check applies. The DMA controller will generate write transactions starting at this address, incrementing by 4 bytes per beat.

DST_ADDR[31:2] — byte address, bits [31:2]	Must be 00[1:0]

Valid range: 0x2000_0000 to 0x2000_3FFC (SRAM1 window, word-aligned).

Why separate src and dst rather than an offset? Because SRAM0 and SRAM1 are at completely different 32-bit base addresses (0x10__ vs 0x20__). Using an offset model would require software to subtract the base first, adding unnecessary complexity. With explicit registers, software programs absolute addresses, which is both simpler and safer.

4.5 DMA_LEN — Transfer Length Register (Offset 0x10)

Specifies how many 32-bit words to transfer. The DMA decrements an internal copy of this value as each word completes. Writing 0 causes immediate DMA_ERR_ZERO_LEN. There is no upper-bound check beyond the SRAM window: software must ensure src_addr + (len × 4) does not exceed 0x1000_3FFF.

TRANSFER_LEN[31:0] — number of 32-bit words

Bit(s)	Name	Access	Description
[31:0]	LEN	R/W	Transfer word count. Practical range: 1 to 4096 for a fully within-SRAM transfer. The DMA does not automatically clamp this — program it carefully.

4.6 Performance Counter Registers (0x14, 0x18, 0x1C)

All three reset to 0 at each START. Stop counting when FSM enters DONE or ERROR. Read-only to software.

These three 32-bit registers are written exclusively by the DMA controller and are read-only to software. They reset to 0 at the start of each new transfer (when START is asserted). They stop counting when the FSM exits the active transfer states (DONE or ERROR).

Register	Offset	What It Counts
DMA_PERF_CYCLES	0x14	Total AHB clock cycles from START pulse to FSM entering DMA_DONE/DMA_ERROR. Divide by HCLK frequency to get wall-clock time.

Register	Offset	What It Counts
DMA_PERF_BEATS	0x18	Successful AHB data beats: incremented each time HREADY=1 in DMA_READ_WAIT or DMA_WRITE_WAIT. For N words: ideal value = 2×N (N reads + N writes).
DMA_PERF_STALLS	0x1C	HREADY=0 cycles: incremented each clock where the FSM is in a _WAIT state and HREADY is still low. Measures bus contention or slow SRAM response.

Performance counter usage example:

```
cycles = read(DMA_PERF_CYCLES);
beats  = read(DMA_PERF_BEATS);   // ideal = 2 × DMA_LEN (1 read + 1 write per
word)
stalls = read(DMA_PERF_STALLS);
bus_efficiency = (float)beats / (float)cycles * 100.0;
// Ideal: 100% means every cycle produced a completed beat (zero stalls)
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5. Error Code Reference

When FSM enters DMA_ERROR, STATUS[7:4] (ERRCODE) latches the cause. Errors 1–3 are pre-flight. Errors 4–6 are runtime.

Code	Enum Name	Meaning & When It Fires
4'd0	DMA_ERR_NONE	No error. Clears at next START.
4'd1	DMA_ERR_ZERO_LEN	DMA_LEN=0. Transfer blocked; DMA stays idle.
4'd2	DMA_ERR_SRC_ALIGN	DMA_SRC_ADDR[1:0] ≠ 00. Must be word-aligned.
4'd3	DMA_ERR_DST_ALIGN	DMA_DST_ADDR[1:0] ≠ 00. Must be word-aligned.
4'd4	DMA_ERR_SLV_RESP	AHB slave returned HRESP=ERROR during a beat.
4'd5	DMA_ERR_ABORT	Software wrote CTRL[2]=1 mid-transfer.
4'd6	DMA_ERR_OUT_OF_RANGE	Address walked outside SRAM window mid-transfer.

6. FSM State Reference

The DMA controller uses a 9-state Moore FSM. Outputs (AHB signals, status register bits) depend only on the current state, not on the input that caused the transition. The state encoding uses 4 bits (dma_state_t typedef) allowing up to 16 states for future expansion.

State	Description
DMA_IDLE	Waits for CTRL[0]=1. Validates alignment + LEN>0 before advancing.
DMA_SETUP	Latches src/dst pointers; runs addr_in_range() checks; clears prev status.
DMA_READ_ADDR	Drives HADDR=src_ptr, HTRANS=NONSEQ (or SEQ in burst), HWRITE=0.
DMA_READ_WAIT	Holds until HREADY=1. Captures HRDATA.
DMA_WRITE_ADDR	Drives HADDR=dst_ptr, HTRANS=NONSEQ, HWRITE=1. HWDATA=latched.
DMA_WRITE_WAIT	Holds until HREADY=1. Increments ptrs +4, decrements counter.
DMA_UPDATE	Counter==0? → DONE. Else → READ_ADDR.
DMA_DONE	Asserts STATUS[DONE]. Pulses IRQ if IRQ_EN=1.
DMA_ERROR	Sets STATUS[ERROR] and ERRCODE. Stays until new START.

State transition summary:

IDLE → SETUP (on valid START), SETUP → READ_ADDR → READ_WAIT → WRITE_ADDR → WRITE_WAIT → UPDATE → (back to READ_ADDR if words remain, else) DONE. Any state → ERROR on abort or bus fault.

6.1 AHB-Lite Timing Diagrams

Single-Beat Transfer (MODE=0, BURST_LEN=1) — One Word Copy

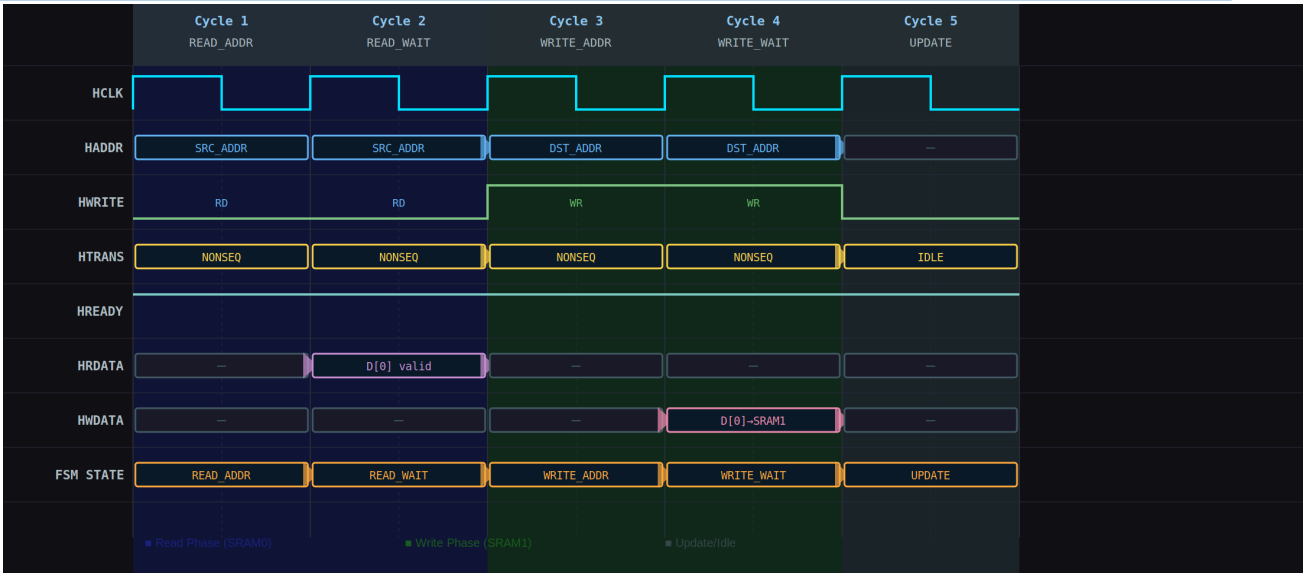

Figure 2 — Single-beat timing: Cycles 1–2 read from SRAM0 (HRDATA valid C2), Cycles 3–4 write to SRAM1 (HWDATA=D[0] on C4), Cycle 5 FSM updates counter.

Note: HRDATA captured when HREADY=1 in READ_WAIT. HWDATA held stable through WRITE_WAIT until HREADY=1 acknowledges the write. For N words, this 5-cycle pattern repeats N times → PERF_BEATS = 2×N (1 read + 1 write per word).

Burst Transfer (MODE=1, BURST_LEN=3) — Three Words in One Burst

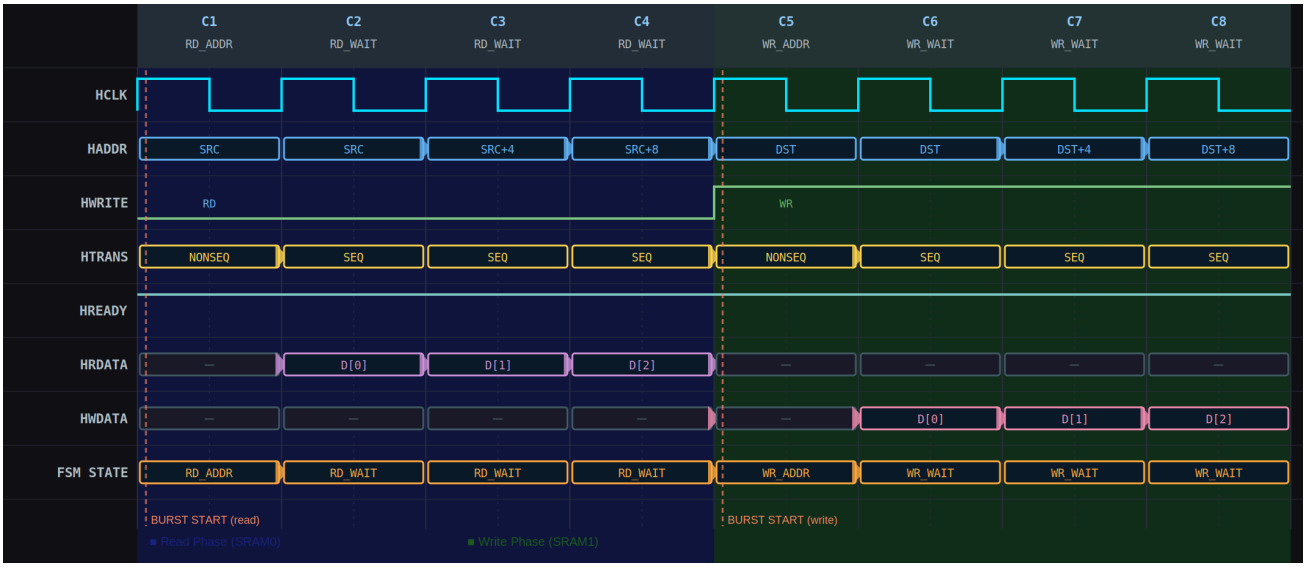

Figure 3 — Burst timing: C1–C4 burst-read 3 words from SRAM0 (NONSEQ then SEQ×2), C5–C8 burst-write to SRAM1. Orange dashes mark burst start boundaries.

Note: First beat of each burst: HTRANS=NONSEQ. Subsequent beats: HTRANS=SEQ. HRDATA is pipelined — valid one cycle after address phase while HREADY=1. HWDATA for each write beat carries the corresponding latched read data.

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7. AHB-Lite Interface Signal Reference

Single master, two slaves. AHB-Lite only: no arbiter, no split transactions, no retry — OKAY or ERROR only.

Signal	Dir	Role in this DMA
HCLK	in	Clock. All FSM transitions and register writes synchronous to rising edge.
HRESETn	in	Active-low reset. Returns FSM to DMA_IDLE; clears all registers.
HADDR	out	32-bit byte address. SRC_ADDR during read phase, DST_ADDR during write phase.
HWRITE	out	0 = read (SRAM0). 1 = write (SRAM1).
HSIZE	out	Always 3'b010 (HSIZE_WORD). 32-bit word transfers only.

Signal	Dir	Role in this DMA
HTRANS	out	IDLE (2'b00) when inactive; NONSEQ (2'b10) first beat; SEQ (2'b11) burst continuation.
HWDATA	out	Data latched from HRDATA during read phase; presented here on following write phase.
HRDATA	in	Read data from SRAM0. Captured when HREADY=1 in READ_WAIT.
HREADY	in	Slave handshake. DMA holds phase stable until this is high.
HRESP	in	1'b1 = slave error → DMA transitions to DMA_ERROR(DMA_ERR_SLV_RESP).

8. Burst Length Encoding

CTRL[7:4] stores burst count directly as 4-bit integer. Maps to AHB INCR burst types when MODE=1.

CTRL[7:4]	Enum	AHB Burst	When to Use
`4'd1`	`BURST_LEN_1`	INCR1 (NONSEQ only)	Default/safe. One beat per word.
`4'd4`	`BURST_LEN_4`	INCR4	Good throughput on verified fabrics.
`4'd8`	`BURST_LEN_8`	INCR8	High efficiency; SRAM must sustain 8 beats.
`4'd16`	`BURST_LEN_16`	INCR16	Max throughput; verify end-to-end.